

Fine TIME

Scientific Executive Summary

2 October 2026

Research question

Fine TIME asks whether task-specific fine-tuning improves pretrained time-series foundation models over their frozen zero-shot forecasts. Each task compares the same backbone before and after adaptation, using the same forecast windows and evaluation support.

Implemented experiment

The first experiment covers Chronos-2, Chronos-Bolt, and TS-ICL on the 90-task Selectime subset used by the project. Each task produces paired frozen and fine-tuned forecasts. Fine-tuning uses only model-eligible training windows with finite input and output support; it does not substitute an unfitted model when no such window exists. Seasonal Naive supplies the shared reference grid and scaling contract.

Component	Current contract
Models	Chronos-2, Chronos-Bolt, and TS-ICL
Comparison	Frozen forecast versus task-fine-tuned forecast
Task set	90-task Selectime subset
Outputs	Paired predictions, metrics, timing, manifests, and report artifacts
Run identity	Meaningful scientific path plus <code>run_n</code> ; configuration and lifecycle in <code>manifest.json</code>

Observed evidence

The first synchronized run completed 90 frozen/fine-tuned task pairs for each backbone. Lower MASE is better.

Model	Frozen	Fine-tuned	Change	Win rate
Chronos-2	1.06923	1.05234	−1.58%	75.56%
Chronos-Bolt	1.16790	1.12307	−3.84%	80.00%
TS-ICL (batch 8)	1.09871	1.09506	−0.33%	81.11%

Fine-tuning improved the equal-task-weighted mean in all three observed arms, most strongly for Chronos-Bolt. The task-level effect was heterogeneous, so these averages do not imply a benefit on every task.

Current boundary

The Chronos-2 and Chronos-Bolt comparisons remain compatible with the current configuration. The observed TS-ICL arm used the former batch size of 8; the current default is 32. Its 90 fine-tuned checkpoints and adapted evaluations must therefore be rerun before a current TS-ICL conclusion is available. Frozen TS-ICL evaluations remain reusable. The existing report also omits the exact selected input-manifest references; that provenance repair requires only report regeneration. Results use one seed and 100 optimization steps, so they do not establish statistical significance or broader-profile generalization.